DRESS and the WL Hierarchy: Climbing One Vertex at a Time

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Abstract

The Cai–Fürer–Immerman (CFI) construction provides the canonical family of hard instances for the Weisfeiler–Leman (WL) hierarchy: distinguishing the two non-isomorphic CFI graphs over a base graph G requires WL of dimension at least the treewidth of G. In this paper, we introduce two iterated extensions of the DRESS continuous structural refinement framework: ∇^k-DRESS, which applies k levels of vertex individualization (weight marking), and Δ^k-DRESS, which applies k levels of vertex deletion. Both enumerate all $\binom{n}{k}$ vertex subsets and collect DRESS fingerprints into a multiset.

We prove that ∇^k-DRESS is at least as powerful as $(k+2)$-WL: the base case shows that Original-DRESS is at least as powerful as 2-WL via the injectivity of the Birkhoff contraction fixed point, and an inductive argument lifts the result by one WL level per individualization depth. We also show that Δ^k-DRESS achieves the same guarantee under the Reconstruction Conjecture, establishing Δ^k-DRESS as a practical variant.

We verify the theoretical bound empirically on the canonical CFI benchmark: Original-DRESS ($k = 0$) already distinguishes CFI(K_3) (requiring 2-WL), and each additional level extends the range by one WL level, up to $k = 3$ matching 5-WL. Crucially, $k = 3$ fails on CFI(K_7) (requiring 6-WL), confirming a sharp boundary at $(k+2)$-WL. Both ∇^k and Δ^k produce identical outcomes on every CFI pair tested. The computational cost is $\mathcal{O}\big(\binom{n}{k} \cdot I \cdot m \cdot d_{\max}\big)$—polynomial in n for fixed k. These results establish iterated DRESS as a practical framework for systematically climbing the WL hierarchy.

1 Introduction

The Weisfeiler–Leman (WL) hierarchy [10] is the standard yardstick for measuring the expressiveness of graph isomorphism algorithms. The j-WL test operates on j-tuples of vertices and runs in $\mathcal{O}(n^{j+1})$ time per iteration [2]. While powerful in theory, even 3-WL at $\mathcal{O}(n^4)$ is impractical for large graphs.

The CFI construction [2] provides the canonical hard instances: for any base graph G, it produces a pair of non-isomorphic graphs CFI(G) and CFI$'(G)$ that j-WL cannot distinguish whenever j is below the treewidth of G. In particular, CFI(K_n) requires $(n-1)$-WL, making these the hardest known instances for the WL hierarchy.

In the companion paper [4], we introduced the DRESS family of continuous structural refinement algorithms. ∇-DRESS, which marks each vertex by applying a multiplicative weight to its incident edges, was shown to empirically distinguish strongly regular graphs that confound 3-WL. Its deletion-based counterpart Δ-DRESS produces identical outcomes and connects to the Kelly–Ulam reconstruction conjecture [8, 9].

In this paper, we introduce ∇^k-DRESS, which applies k levels of vertex *individualization* (marking the selected vertices by applying a multiplicative weight $\alpha \neq 1$ to their incident edges). This is the natural continuous analogue of the classical individualization operator that defines the WL hierarchy: $(j+1)$-WL is equivalent to running j-WL over all single-vertex individualizations [2, 6].

We also retain Δ^k-DRESS (vertex deletion) as a practical variant: it is empirically indistinguishable from ∇^k-DRESS on every CFI pair we tested, and avoids the need for a weight

parameter. We verify both variants on the CFI family: $k = 0$ (Original-DRESS) already distinguishes CFI(K_3), $k = 1$ reaches CFI(K_4), $k = 2$ reaches CFI(K_5), and $k = 3$ reaches CFI(K_6).

1.1 Contributions

1. We define ∇^k-DRESS (individualization) and Δ^k-DRESS (deletion) formally and analyze their complexity.
2. We prove that ∇^k-DRESS is at least as powerful as $(k+2)$-WL: Original-DRESS matches 2-WL via injectivity of the Birkhoff contraction fixed point, and each individualization level adds one WL level by induction (Theorems 6 and 9).
3. We show that Δ^k-DRESS is an empirically equivalent practical variant that avoids the weight parameter, and is provably equivalent under the Reconstruction Conjecture.
4. We verify the theoretical bound empirically on the canonical CFI benchmark family, showing that ∇^k-DRESS distinguishes CFI(K_{k+3}) for each $k = 0, 1, 2, 3$, and fails at the next level, covering WL levels 2 through 5. Δ^k-DRESS produces identical outcomes on all tested pairs.
5. We observe a sharp boundary at $(k + 2)$-WL: $k = 3$ fails on CFI(K_7) (requiring 6-WL).
6. We discuss the theoretical implications and open questions raised by these results.

2 Preliminaries

2.1 DRESS

The Original-DRESS equation [3] is a parameter-free, non-linear dynamical system on edges:

$$d_{uv}^{(t+1)} = \frac{\sum_{x \in N[u] \cap N[v]} \left(d_{ux}^{(t)} + d_{xv}^{(t)}\right)}{\|u\|^{(t)} \cdot \|v\|^{(t)}} \tag{1}$$

where $\|u\|^{(t)} = \sqrt{\sum_{x \in N[u]} d_{ux}^{(t)}}$ is the vertex norm and $N[u] = N(u) \cup \{u\}$ is the closed neighborhood. Self-loops are added to every vertex; the invariant $d_{uu} = 2$ holds at every iteration. The equation converges to a unique fixed point $d^* \in [0, 2]^{|E|}$ for any positive initialization [4].

The graph fingerprint can be represented equivalently as the sorted vector $\mathrm{sort}(d^*)$ or as the histogram $h(d^*)$; both uniquely identify the multiset of converged edge values. Since DRESS values are bounded in $[0, 2]$ and convergence tolerance is $\epsilon = 10^{-6}$, each edge value maps to one of $\lceil 2/\epsilon \rceil = 2 \times 10^6$ integer bins (the bin count is determined by the convergence tolerance), so the histogram is a compact bin-count vector. In the experiments we use the histogram representation for efficiency; in the proofs we work with the full multiset fingerprint. Two graphs are declared non-isomorphic if their fingerprints differ.

2.2 ∇-DRESS

∇-DRESS [4] breaks symmetry by *marking* a vertex: applying a distinguished multiplicative weight $\alpha \neq 1$ to its incident edges, while preserving the full graph topology. Given a graph $G = (V, E, w)$ with symmetric weight function $w : E \to \mathbb{R}_{>0}$ and a vertex v, the individualized graph $G^{(\alpha,v)}$ has weights:

$$w^{(\alpha,v)}(e) = \begin{cases} \alpha \cdot w(e) & \text{if } v \in e, \\ w(e) & \text{otherwise.} \end{cases}$$

The ∇-DRESS fingerprint is the multiset of per-vertex DRESS fingerprints:

$$\nabla\text{-DRESS}(\mathcal{F}, G, \alpha) = \{\{\text{sort}(\mathcal{F}(G^{(\alpha,v)})) : v \in V\}\}$$

Equivalently, a pooled histogram can be formed by accumulating all converged edge values across all n individualizations.

2.3 Δ-DRESS

Δ-DRESS [4] runs DRESS on each vertex-deleted subgraph:

Definition 1 (Δ-DRESS)**.** *Given a graph $G = (V, E)$ and a DRESS variant $\mathcal{F}$, the Δ-DRESS fingerprint is the histogram obtained by running $\mathcal{F}$ on each vertex-deleted subgraph and accumulating every edge similarity value into a single bin-count vector:*

$$\Delta\text{-}DRESS(\mathcal{F}, G) = h\left(\bigsqcup_{v \in V} \mathcal{F}(G \setminus \{v\})\right)$$

where $G \setminus \{v\}$ is the subgraph induced by $V \setminus \{v\}$, $\bigsqcup$ denotes multiset union, and h maps each value to its integer bin in a fixed-size array.

3 ∇^k-DRESS and Δ^k-DRESS

We now define the iterated individualization operator and its deletion-based counterpart.

Definition 2 (∇^k-DRESS — Individualization)**.** *For a graph $G = (V, E, w)$ with symmetric edge weights $w : E \to \mathbb{R}_{>0}$, a DRESS variant $\mathcal{F}$, an individualization weight $\alpha \neq 1$, and an individualization depth $k \geq 0$:*

$$\nabla^k\text{-}DRESS(\mathcal{F}, G, \alpha) = h\left(\bigsqcup_{S \subset V,\ |S|=k} \mathcal{F}(G^{(\alpha,S)})\right)$$

where $G^{(\alpha,S)}$ is obtained from G by multiplying the weight of every edge incident to at least one vertex in S by α:

$$w^{(\alpha,S)}(e) = \begin{cases} \alpha \cdot w(e) & \text{if } e \cap S \neq \emptyset, \\ w(e) & \text{otherwise.} \end{cases}$$

The histogram h pools edge values across all $\binom{n}{k}$ individualized runs.

Individualization preserves the full graph topology: all edges remain present, with marked edges receiving a distinguished weight. This makes ∇^k-DRESS the natural continuous analogue of the classical individualization operator that defines the WL hierarchy. At depth k, ∇^k-DRESS computes DRESS on all $\binom{n}{k}$ weight-modified copies of G. Every individual edge similarity value from every run is added to a single histogram. Two graphs are compared by checking equality of their histograms. The memory footprint is constant regardless of k, whereas storing the raw multiset of $\binom{n}{k} \cdot |E|$ floating-point values would be prohibitive.

Definition 3 (Δ^k-DRESS — Deletion variant)**.** *For a graph $G = (V, E)$, a DRESS variant $\mathcal{F}$, and a deletion depth $k \geq 0$:*

$$\Delta^k\text{-}DRESS(\mathcal{F}, G) = h\left(\bigsqcup_{S \subset V,\ |S|=k} \mathcal{F}(G \setminus S)\right)$$

where $G \setminus S$ is the subgraph induced by $V \setminus S$, and h maps the pooled edge values into a histogram.

Proposition 4 (Complexity)**.** *The time complexity of both ∇^k-DRESS and Δ^k-DRESS is*

$$\mathcal{O}\left(\binom{n}{k} \cdot I \cdot m \cdot d_{\max}\right)$$

where $n = |V|$, $m = |E|$, $d_{\max}$ is the maximum degree, and I is the number of DRESS iterations per subgraph. The $\binom{n}{k}$ computations are entirely independent and embarrassingly parallel. ∇^k-DRESS has a small constant-factor advantage: it requires only $\mathcal{O}(d_{\max} \cdot k)$ weight modifications per subset, operating on the full graph without rebuilding.

Comparison with j-WL. The cost of j-WL is $\mathcal{O}(n^{j+1})$ per iteration. For CFI(K_6), 5-WL costs $\mathcal{O}(n^6)$, while Δ^3-DRESS and ∇^3-DRESS cost $\mathcal{O}(n^3 \cdot I \cdot m \cdot d_{\max}) = \mathcal{O}(n^3 \cdot m \cdot d_{\max})$ (with I constant).

4 Expressiveness Relative to the Weisfeiler–Lehman Hierarchy

We now prove that ∇^k-DRESS is at least as powerful as the $(k+2)$-dimensional Weisfeiler–Lehman ($(k+2)$-WL) test. We first establish the base case, then lift the result by induction on the individualization depth. We subsequently discuss Δ^k-DRESS as a practical variant.

We first establish a key property of the DRESS fixed-point operator.

Lemma 5 (Refinement monotonicity)**.** *Let F denote the Original-DRESS update operator, and let $\mathbf{d}^{(t)}$ denote the edge-value vector at iteration t. If two edges (u, v) and (u', v') have the same value $d_{uv}^{(t)} = d_{u'v'}^{(t)}$ at iteration t but differ in their common-neighbor multiset structure—i.e., the multisets $\{\{(d_{ux}^{(t)}, d_{xv}^{(t)}) : x \in N[u] \cap N[v]\}\}$ and $\{\{(d_{u'x'}^{(t)}, d_{x'v'}^{(t)}) : x' \in N[u'] \cap N[v']\}\}$ are not equal—then $d_{uv}^{(t+1)} \neq d_{u'v'}^{(t+1)}$.*

Proof. The DRESS update for edge (u, v) is:

$$d_{uv}^{(t+1)} = \frac{\sum_{x \in N[u] \cap N[v]} (d_{ux}^{(t)} + d_{xv}^{(t)})}{\|u\|^{(t)} \cdot \|v\|^{(t)}}$$

where $\|u\|^{(t)} = \sqrt{\sum_{x \in N[u]} d_{ux}^{(t)}}$. This is a rational function of the input values. At each iteration, the numerator is a sum over the common-neighbor multiset $N[u] \cap N[v]$, and the denominator depends on the full closed neighborhoods $N[u]$ and $N[v]$.

Suppose, for contradiction, that two edges with distinct common-neighbor multiset structures always produce the same output. This would require the rational equation

$$\frac{\sum_{x \in N[u] \cap N[v]} (d_{ux} + d_{xv})}{\|u\| \cdot \|v\|} = \frac{\sum_{x' \in N[u'] \cap N[v']} (d_{u'x'} + d_{x'v'})}{\|u'\| \cdot \|v'\|}$$

to hold identically for all positive $\mathbf{d}$, despite the two sides being structurally different rational functions (aggregating over different-sized or differently-weighted multisets). Since distinct multiset structures yield distinct polynomial expressions in the numerator and/or denominator, equality cannot hold on an open set of positive vectors. In particular, the unique fixed point $\mathbf{d}^*$ (which lies in the interior of the positive cone by the Birkhoff contraction theorem [4]) is a generic point of this open set, so the two values must differ at $\mathbf{d}^*$. □

Theorem 6 (Original-DRESS is at least as powerful as 2-WL)**.** *Let G and H be two graphs. If the stable colorings produced by the 2-dimensional Weisfeiler–Lehman test on G and H differ, then the DRESS fingerprints of G and H differ (i.e., $\mathrm{sort}(\mathbf{d}_G^*) \neq \mathrm{sort}(\mathbf{d}_H^*)$ or $h(\mathbf{d}_G^*) \neq h(\mathbf{d}_H^*)$).*

Proof. The stable 2-WL assigns the same color to two pairs (u, v) and (u', v') if and only if they have identical multisets of common-neighbor colors.

Consider the fixed-point equation of Original-DRESS (triangle motif):

$$d^*_{uv} \cdot \|u\| \cdot \|v\| = \sum_{x \in N[u] \cap N[v]} (d^*_{ux} + d^*_{xv}). \tag{2}$$

The right-hand side aggregates over precisely the same common-neighbor multiset that 2-WL uses for refinement.

We show that DRESS refines at least as finely as 2-WL. The 2-WL coloring is computed by iterated multiset refinement: at each round, pairs with distinct common-neighbor color multisets receive distinct colors. DRESS performs an analogous iteration in continuous space: at each step, the value $d^{(t+1)}_{uv}$ is a function of the multiset $\{\{(d^{(t)}_{ux}, d^{(t)}_{xv}) : x \in N[u] \cap N[v]\}\}$. By Lemma 5, whenever two edges have distinct common-neighbor value-multisets, they receive distinct updated values. Since DRESS starts from a uniform initialization (all edges equal) and 2-WL starts from a uniform coloring (all pairs same color), an easy induction on the iteration count shows that:

If 2-WL assigns different colors to (u, v) and (u', v') at round t, then $d^{(t)}_{uv} \neq d^{(t)}_{u'v'}$.

At the stable coloring (the fixed point of both processes), any color difference in 2-WL is therefore reflected as a value difference in $\mathbf{d}^*$. In particular, if the 2-WL stable colorings of G and H differ—meaning the multisets of stable colors are not equal—then the multisets of fixed-point edge values $\mathbf{d}^*_G$ and $\mathbf{d}^*_H$ are not equal either, and hence $\mathrm{sort}(\mathbf{d}^*_G) \neq \mathrm{sort}(\mathbf{d}^*_H)$. □

Before stating the main inductive theorem, we recall a classical characterization of the WL hierarchy via vertex individualization.

Lemma 7 (WL and vertex individualization)**.** *Let $j \geq 2$ and let G, H be two graphs. If $(j+1)$-WL distinguishes G from H, then the multisets of j-WL colorings over all single-vertex individualizations differ:*

$$\{\{j\text{-}WL(G, v) : v \in V(G)\}\} \neq \{\{j\text{-}WL(H, w) : w \in V(H)\}\}$$

where (G, v) denotes the graph G with vertex v individualized (given a unique color).

Proof. This is a consequence of the characterization of j-WL via the counting logic C^j [2, 7]. The logic C^{j+1} can express "there exists a vertex v such that $\varphi(v)$ holds," where φ is a C^j formula. Since $(j+1)$-WL captures exactly C^{j+1} [2], any distinction made by $(j+1)$-WL can be witnessed by individualizing one vertex and applying j-WL. This is formalized in [6, Theorem 5.1]; see also [5, Section 2] for a self-contained treatment. □

We now define the **multiset fingerprint** of ∇^k-DRESS, which retains full per-individualization information:

Definition 8 (Multiset fingerprint — Individualization)**.** *For a graph $G = (V, E, w)$, a DRESS variant $\mathcal{F}$, an individualization weight $\alpha \neq 1$, and depth $k \geq 0$:*

$$\nabla^k\text{-}DRESS_{\mathrm{ms}}(\mathcal{F}, G, \alpha) = \{\{\mathrm{sort}(\mathcal{F}(G^{(\alpha,S)})) : S \subset V,\ |S| = k\}\}$$

i.e., the multiset of per-individualization sorted fingerprint vectors. This is strictly more informative than the pooled histogram of Definition 2.

Theorem 9 (∇^k-DRESS is at least as powerful as $(k+2)$-WL)**.** *For any fixed integer $k \geq 0$ and any individualization weight $\alpha \neq 1$, if the $(k+2)$-dimensional Weisfeiler–Lehman test distinguishes two graphs G and H, then their ∇^k-DRESS multiset fingerprints differ:*

$$\nabla^k\text{-}DRESS_{\mathrm{ms}}(\mathcal{F}, G, \alpha) \neq \nabla^k\text{-}DRESS_{\mathrm{ms}}(\mathcal{F}, H, \alpha).$$

Proof. We proceed by induction on k.

Base case ($k = 0$): ∇^0-DRESS reduces to Original-DRESS (no vertices are individualized, and the empty subset $S = \emptyset$ contributes one run). The multiset fingerprint is $\{\{\text{sort}(\mathbf{d}^*_G)\}\}$. The claim is exactly Theorem 6.

Inductive hypothesis: Assume the statement holds for depth $k-1$: whenever $(k+1)$-WL distinguishes two graphs, their ∇^{k-1}-DRESS multiset fingerprints differ.

Inductive step: Let $(k+2)$-WL distinguish G and H. By Lemma 7, the multisets of $(k+1)$-WL colorings over all single-vertex individualizations differ:

$$\{\{(k{+}1)\text{-WL}(G, v) : v \in V(G)\}\} \neq \{\{(k{+}1)\text{-WL}(H, w) : w \in V(H)\}\}.$$

Now consider the ∇^k-DRESS multiset fingerprint. Every subset $S \subset V(G)$ with $|S| = k$ can be decomposed as $S = \{v\} \cup S'$ where $|S'| = k-1$. The weight modification $G^{(\alpha,S)}$ applies α to all edges incident to any vertex in S. This is equivalent to first individualizing v (applying α to edges incident to v) and then individualizing S' in the resulting graph: $G^{(\alpha,S)} = (G^{(\alpha,\{v\})})^{(\alpha,S')}$, since the multiplicative modification is applied independently to each vertex in S.

Collecting over all $(k-1)$-subsets S' for a fixed v, this is precisely ∇^{k-1}-DRESS applied to the individualized graph $G^{(\alpha,\{v\})}$. Therefore:

$$\nabla^k\text{-DRESS}_{\text{ms}}(\mathcal{F}, G, \alpha) = \biguplus_{v \in V(G)} \nabla^{k-1}\text{-DRESS}_{\text{ms}}(\mathcal{F}, G^{(\alpha,\{v\})}, \alpha).$$

Now, vertex individualization in ∇-DRESS (multiplying incident edge weights by α) is a continuous analogue of the discrete WL individualization (assigning a unique color). By the same reasoning as Theorem 6—the DRESS fixed point is injective on structures that WL distinguishes—if $(k{+}1)$-WL distinguishes (G, v) from (H, w) for the corresponding vertex matchings, then ∇^{k-1}-DRESS on $G^{(\alpha,\{v\})}$ differs from ∇^{k-1}-DRESS on $H^{(\alpha,\{w\})}$ by the inductive hypothesis.

Since the multisets of $(k+1)$-WL individualizations differ between G and H, the multisets of ∇^{k-1}-DRESS fingerprints over all single-vertex individualizations differ. This difference is preserved in the multiset union, completing the induction. □

Remark 10. *The proof is stated for the triangle-based Original-DRESS update. It extends verbatim to any Motif-DRESS variant (with the base case becoming $\geq (r+1)$-WL when the motif M has size $r+1$). The individualization mechanism that climbs the hierarchy remains unchanged.*

4.1 Δ^k-DRESS as a Practical Variant

We now relate Δ^k-DRESS to ∇^k-DRESS. While deletion and individualization are distinct operations, they are closely connected.

Definition 11 (Multiset fingerprint — Deletion)**.** *For a graph $G = (V, E)$, a DRESS variant $\mathcal{F}$, and a deletion depth $k \geq 0$:*

$$\Delta^k\text{-}DRESS_{\text{ms}}(\mathcal{F}, G) = \{\{\text{sort}(\mathcal{F}(G \setminus S)) : S \subset V,\ |S| = k\}\}$$

Remark 12 (Individualization vs. Deletion)**.** *Vertex individualization is at least as informative as vertex deletion: given $G^{(\alpha,\{v\})}$ (all edges present, marked edges distinguished), one can recover the structure of $G \setminus \{v\}$ by examining only the unmarked edges. The converse does not hold in general: the vertex-deleted subgraph $G \setminus \{v\}$ loses all information about the edges incident to v and how they connect to the rest of the graph.*

The Reconstruction Conjecture [8, 9] asserts that for $n \geq 3$, a graph is determined (up to isomorphism) by its deck of n vertex-deleted subgraphs. If this conjecture holds, then any graph property distinguishable via individualization is also distinguishable via deletion, and the two operators become equivalent in discriminative power.

Corollary 13 (Δ^k-DRESS under the Reconstruction Conjecture)**.** *If the Reconstruction Conjecture holds, then for any $k \geq 0$, Δ^k-DRESS is at least as powerful as $(k+2)$-WL:*

$$\Delta^k\text{-}DRESS_{\text{ms}}(\mathcal{F}, G) \neq \Delta^k\text{-}DRESS_{\text{ms}}(\mathcal{F}, H)$$

whenever $(k+2)$-WL distinguishes G from H.

Proof. Under the Reconstruction Conjecture, if the multisets of individualized fingerprints differ (Theorem 9), then the multisets of deletion fingerprints also differ, since deletion is recoverable from the individualized graph. The result follows by combining Theorem 9 with Remark 12. □

Remark 14 (Histogram representation)**.** *Both ∇^k and Δ^k-DRESS use pooled histograms in practice. The multiset fingerprint (which retains per-subset identity) is strictly more informative and provides the guarantee. The pooled histogram is a coarsening: different multisets could in principle produce the same histogram. Nevertheless, the pooled histogram suffices empirically for all CFI instances tested (Table 1).*

5 Empirical Results

All experiments use Original-DRESS as the base variant $\mathcal{F}$, with convergence tolerance $\epsilon = 10^{-6}$ and a maximum of 100 iterations. In every case, convergence was reached before the iteration limit.

5.1 CFI Graphs

The CFI construction [2] is defined over a base graph $G = (V_G, E_G)$. For each vertex $v \in V_G$, a "gadget" replaces v with a set of vertices encoding its incident edges. Two non-isomorphic graphs CFI(G) and CFI$'(G)$ are produced by twisting the parity of one gadget. The key property is:

Theorem 15 (Cai, Fürer, Immerman [2])**.** *The j-WL algorithm cannot distinguish CFI(G) from CFI$'(G)$ whenever $j < tw(G)$, where $tw(G)$ is the treewidth of G.*

For the complete graph K_n, the treewidth is $n-1$, so CFI(K_n) requires $(n-1)$-WL. In particular:

- CFI(K_3): requires 2-WL.
- CFI(K_5): requires 4-WL, operating on 4-tuples—$\mathcal{O}(n^5)$ per iteration.
- CFI(K_6): requires 5-WL, operating on 5-tuples—$\mathcal{O}(n^6)$ per iteration.

5.2 ∇^k-DRESS and Δ^k-DRESS on CFI(K_n)

We test ∇^k-DRESS (with $\alpha = 2$) and Δ^k-DRESS for $k = 0, 1, 2, 3$ on CFI pairs over complete graphs K_n ($n = 3, \ldots, 10$). Both variants produce identical outcomes (distinguish/fail) on every pair tested. Higher depths on larger base graphs were not executed due to time constraints (marked — in the table).

5.3 Analysis

The results in Table 1 reveal a clear staircase pattern:

1. ∇^0 **(Original-DRESS)** distinguishes CFI(K_3), which requires 2-WL. This is a notable result: Original-DRESS—with no individualization—already exceeds 1-WL on CFI instances, consistent with the prism vs. $K_{3,3}$ result in [4]. It fails on all larger CFI pairs.

Table 1: ∇^k-DRESS results on CFI graph pairs (Δ^k-DRESS produces identical outcomes on all tested pairs). ✓ = pair distinguished; × = pair not distinguished; — = not executed due to time constraints. The WL requirement column shows the minimum WL dimension needed to distinguish the pair.

Base graph	$\lvert V_{\text{CFI}}\rvert$	WL req.	∇^0	∇^1	∇^2	∇^3
K_3	6	2-WL	✓	✓	✓	✓
K_4	16	3-WL	×	✓	✓	✓
K_5	40	4-WL	×	×	✓	✓
K_6	96	5-WL	×	×	×	✓
K_7	224	6-WL	×	×	×	×
K_8	512	7-WL	×	×	—	—
K_9	1152	8-WL	×	×	—	—
K_{10}	2560	9-WL	×	×	—	—

2. ∇^1 extends the range to CFI(K_4) (3-WL). This is consistent with the results in [4], where ∇-DRESS was shown to distinguish SRG pairs that confound 3-WL. It fails on CFI(K_5).
3. ∇^2 reaches CFI(K_5) (4-WL).
4. ∇^3 reaches CFI(K_6) (5-WL) but fails on CFI(K_7) (6-WL), confirming that the staircase has a sharp boundary at $(k+2)$-WL. The CFI(K_7) test required processing all $\binom{224}{3}$ = 1,848,224 subgraphs (≈19.3 billion edge values).

WL level per individualization depth. The tested data shows:

Individualization depth k	Max WL matched	Effective WL
0	2-WL	$k+2$
1	3-WL	$k+2$
2	4-WL	$k+2$
3	5-WL	$k+2$

The pattern is strikingly regular: ∇^k-DRESS empirically matches $(k+2)$-WL on the CFI family, consistent with the lower bound proved in Theorem 9. Δ^k-DRESS produces identical outcomes on all tested pairs (Corollary 13 provides the theoretical guarantee under RC). Each level adds one WL level of expressiveness, with the base DRESS contributing 2-WL on its own. The computational cost grows as $\binom{n}{k}$ (polynomial in n for fixed k), while the equivalent $(k+2)$-WL cost grows as n^{k+3}.

6 Discussion

6.1 Relationship to Existing Work

Subgraph GNNs. Methods such as ESAN [1] and GNN-AK+ [11] also use vertex-deleted or vertex-marked subgraphs to boost expressiveness. However, these are supervised methods that learn aggregation functions from data. ∇^k-DRESS and Δ^k-DRESS are entirely unsupervised: the aggregation is the deterministic DRESS fixed point, and the histogram comparison is parameter-free. ∇^k-DRESS is the closest continuous analogue of the vertex-marking approach used in subgraph GNNs, while Δ^k-DRESS corresponds to vertex deletion.

Reconstruction conjecture. The Kelly–Ulam conjecture [8, 9] states that graphs with $n \geq 3$ are determined by their deck of vertex-deleted subgraphs. Conceptually, Δ^1-DRESS computes a continuous relaxation of this deck: each "card" is the DRESS fingerprint of a vertex-deleted subgraph. The introduction of ∇^k-DRESS is significant because the individualization-based proof (Theorem 9) requires no assumptions beyond the standard DRESS convergence guarantee, while the deletion-based guarantee (Corollary 13) requires RC. The empirical equivalence of both variants on all tested instances suggests that RC holds for the relevant properties, but this remains an open question in general.

Fixed-parameter tractability. For fixed k, both ∇^k-DRESS and Δ^k-DRESS run in polynomial time. Theorem 9 guarantees that ∇^k-DRESS is at least as powerful as $(k+2)$-WL, and the CFI experiments confirm this bound is tight: the minimum depth needed to distinguish CFI(K_n) grows as $n-2$, which is linear in n. While each additional level is polynomial, the number of required levels grows with the base graph size.

6.2 Open Questions

1. **Does the staircase pattern continue beyond $k = 3$?** Theorem 9 establishes $\geq (k+2)$-WL. We have confirmed empirically that $k = 3$ fails on CFI(K_7) (6-WL), so the bound is tight for $k \leq 3$. Testing $k = 4$ on CFI(K_7) would confirm whether the staircase extends.

2. **Is the tightness specific to CFI, or does it hold more broadly?** The lower bound $(k+2)$-WL is general, but the upper bound may depend on the graph family.

3. **Are there graph families where ∇^k-DRESS (or Δ^k-DRESS) fails for all constant k?** CFI graphs over graphs of unbounded treewidth would be natural candidates.

4. **Does combining iterated individualization with other DRESS variants yield further gains?** The DRESS family includes Motif-DRESS and Generalized-DRESS [4]. Whether a stronger base variant $\mathcal{F}$ shifts the staircase upward or merely overlaps with the individualization/deletion mechanism is an open question.

5. **Can ∇^k and Δ^k be separated?** Our experiments show identical outcomes for all tested CFI instances. It is an open question whether there exist graph pairs where ∇^k-DRESS succeeds but Δ^k-DRESS fails, which would provide a concrete counterexample to a restricted form of the Reconstruction Conjecture.

7 Conclusion

We introduced ∇^k-DRESS, which applies k levels of iterated vertex individualization (multiplicative weight marking) to the DRESS framework, and its deletion-based counterpart Δ^k-DRESS. We proved that ∇^k-DRESS is at least as powerful as $(k+2)$-WL (Theorem 9), and showed that Δ^k-DRESS achieves the same guarantee under the Reconstruction Conjecture (Corollary 13). ∇^k-DRESS was verified empirically on the canonical CFI benchmark family: at $k = 0$, Original-DRESS already distinguishes CFI(K_3), and each additional level adds one WL level. $k = 3$ reaches 5-WL but fails on 6-WL, confirming the bound is tight on CFI instances. Δ^k-DRESS produces identical outcomes on every tested pair. These results establish iterated DRESS as a systematic, scalable method for climbing the WL hierarchy with provable expressiveness guarantees. An open-source implementation is available at `https://github.com/velicast/dress-graph`.

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